

The Man in the Middle

An In-Depth Analysis of Cutoff Decision-Making in Baseball

Team 97

0. Abstract

Cutoff plays are a critical but under-analyzed component of baseball. In this project, we develop a framework to assess cutoff play decision-making using MiLB tracking data. First, we construct a logistic regression model to estimate baserunner safe probabilities for common extra base advancement paths. Next, we combine these probabilities with RE24 run expectancy values to compute expected run values for each possible cutoff action, identifying the optimal cutoff choice for every play. Using these labels, we train a Random Forest classifier using features such as runner and fielder distances, arm strength, and sprint speed, reaching an accuracy of 77% in predicting optimal actions. Our findings reveal that cutoff men overwhelmingly favor aggressive throws despite our model often recommending to hold on to the ball, leading to measurable run expectancy losses. Finally, we present an interactive dashboard that provides team- and player-level cutoff decision-making insights that can supplement scouting and coaching. This project offers a scalable approach to cutoff play evaluation that can be used by professional teams to improve their defensive strategy in crucial situations.

1. Introduction

The bases are loaded with two outs in the bottom of the 8th inning of a two-run ballgame. The batter slices a base hit, bouncing once to the left fielder. The runner on third scores easily and the ambitious runner from second rounds third and darts for home. The throw home is a bullet, low and on target. The crowd anticipates a close play at the plate...but the pitcher cuts the ball off. The run scores, and a sigh of relief is shared among the UC San Diego fans. Meanwhile, we all sat wondering, "Should the pitcher really have done that?"



Click on the image to see the play. UC San Diego's Anthony Potestio hits a 2-run single into left field against UC Davis. The pitcher curiously cuts the throw off from left field, allowing the 2nd runner, Emiliano Gonzalez to score.

[Link to play.](#)

In-depth analyses of cutoff plays are few and far between. Infielders are taught how to line up and where to throw, but do they make the right cutoff decision when it matters? Is it better to cut the outfielder's "hero throw" or let it fly?

In this project, we quantified the efforts of the relay man in order to evaluate cutoff decision-making. We discuss our methodology, the challenges we faced, and how we overcame them. We then showcase an application that stores our insights in an interactive dashboard and close with big-picture takeaways.

2. Data

The data was provided to us by SMT, and can essentially be thought of as an extremely detailed baseball scorebook. Key tables included game info (which team was hitting, which runners were on base), game events (each play broken into events like pitch, ball contact, catch, etc.), ball position (where the ball was over the course of a play), and player position (where each player was located over the course of a play). Importantly, the data only captured the physical events that happened on the field, like pitches, throws, and ball trajectories. It did not include play outcomes like outs, runs scored, errors, and other official scoring results.

2.1 Filtering

We confined our analysis to base hits to the outfield, with runners on first and/or second base. These play types opened the door for potential cutoffs, where the relay man would have to choose between continuing the assist throw, cutting and relaying to another base, or holding the ball. Using this criteria, we ended up with around 1500 unique baserunners over 800 cutoff plays.

2.2 Constraints on the Data

The anonymized data omitted two necessary components: whether runners reached their target base, and the number of outs. We created several conditionals and boundaries for the runner, fielder, and ball coordinates in order to label the baserunner safe or out (see *Appendix 1*). For instance, if the runner was much closer to the base than the infielder when the infielder caught the ball, then we labeled them safe. Additionally, we calculated the number of outs using base occupancy information (see *Appendix 2*). Although these definitions are not perfect, we tested and manually checked enough plays to feel confident in our results.

3. Methodology

Our overall methodology consisted of three main steps: modeling baserunner safe probabilities, calculating optimal cutoff actions, and modeling optimal cutoff decisions. To model the optimal cutoff decision for a given play, we first needed to establish what the optimal cutoff actions were. We did this by combining baserunning safe probabilities with RE24¹ run expectancies.

3.1 Safe Probability Model

To find the safe probabilities, we built a model to predict the probability of a given base runner advancing at least one extra base (1st to 3rd, 2nd to Home, etc.). In the UC San Diego example, our model would produce the probability of Gonzalez reaching home from second at the moment the left fielder fielded the ball.

We used a set of five logistic regression classification models, considering the outfielder's and the baserunner's respective distances from the target base at the time the fielder acquired the ball (*Appendix 5*). We trained five individual models, each considering a different baserunner advancement scenario: 1B → 3B, 1B → home, 2B → home, home → 2B, and home → 3B.

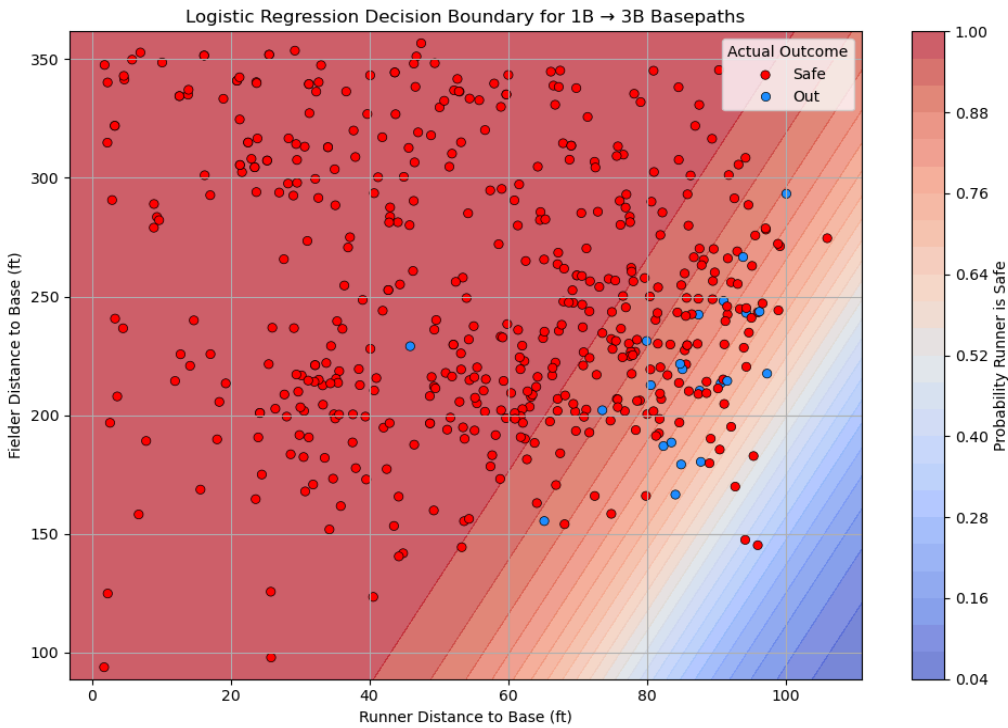


Figure 1 showcases the two features we used and how they inform safe or out labels in our safe probability model, specifically for the 1B → 3B basepath. It displays how the probability the runner is safe changes as the fielder and runner distances to the targeted base change.

¹RE24 - Data representing the expected, or average, run value of every base-out permutation. For example, with 1 out and a runner on first, the expected amount of runs a team will score in that inning is 0.404.

All five models performed well, with brier scores below 0.1:

Safe Probability Model Brier Score by Basepath

BASEPATH	BRIER SCORE
Home → 2B	0.017
1B → 3B	0.028
Home → 3B	0.036
2B → Home	0.054
1B → Home	0.091

Figure 2: Brier scores for each basepath. A Brier score is a metric to evaluate the accuracy of probabilistic predictions. 0 means the probabilities predict the outcome perfectly, and 1 means they never predict the outcome correctly. The lower the Brier score, the better the model.

The safe probability distribution is definitely left skewed, showcased in *Figure 3*. This is because we only considered plays where the runner actually attempted to take an extra base, which inherently skews the distribution as the runner would rarely ever attempt a low-probability extra base. The example play in *figure 4* demonstrates the capability of this initial safe probability model.

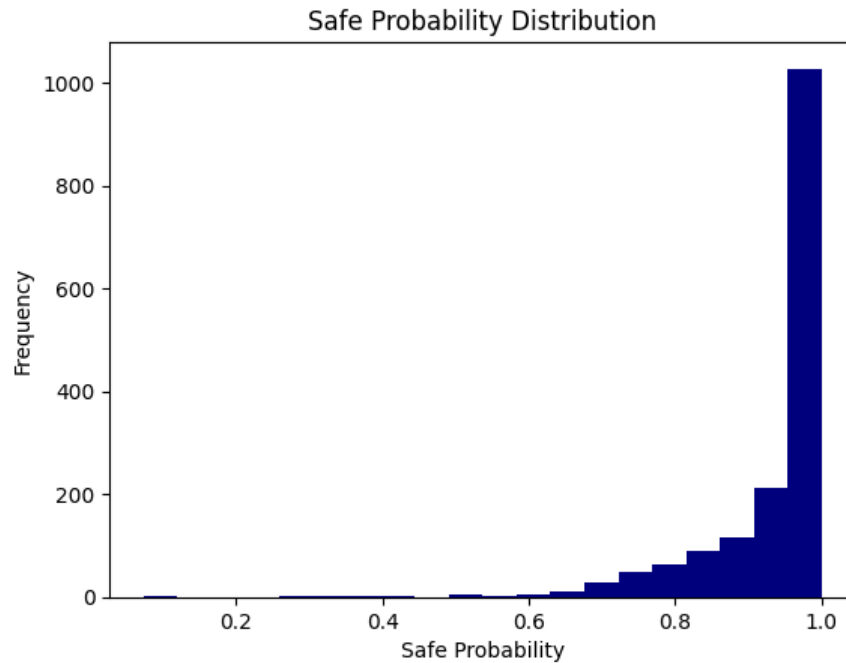


Figure 3: The model's safe probability distribution showcased as a histogram.

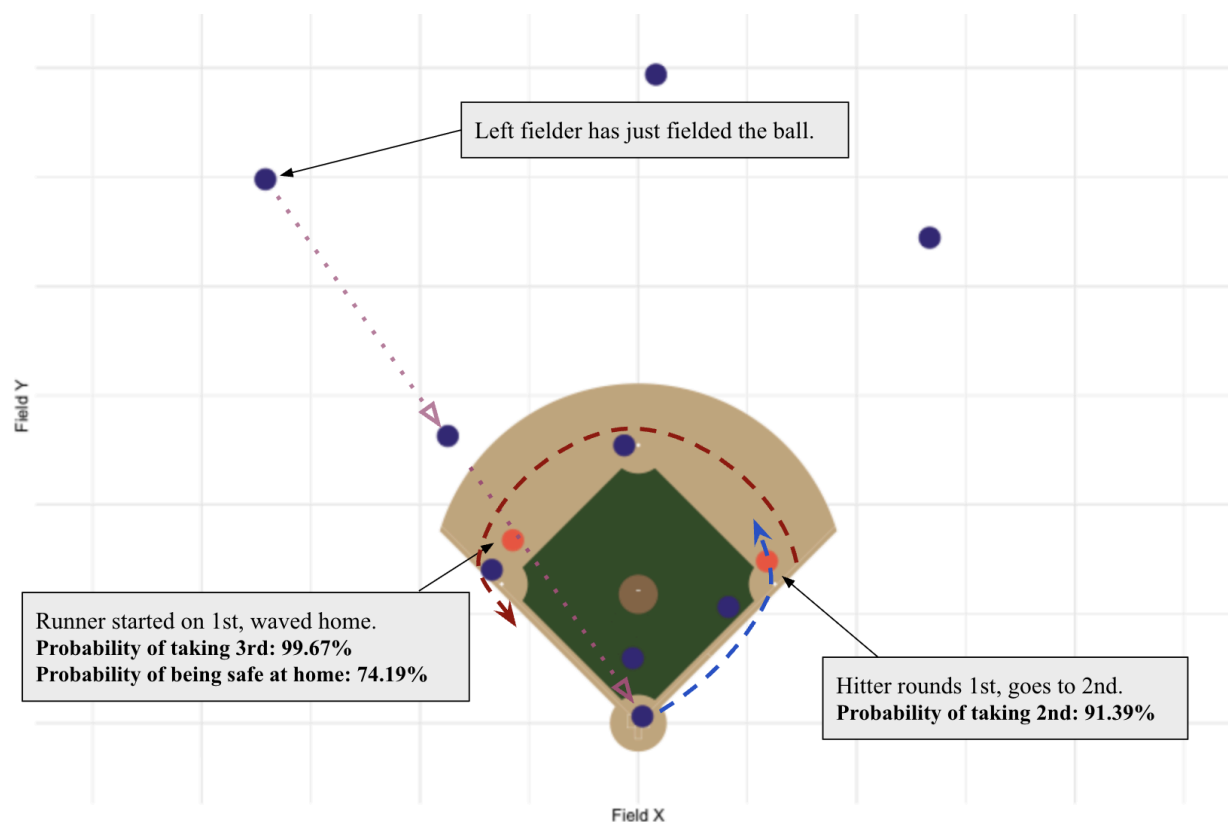


Figure 4: **Click** on the image to see an animation of the play. Showcased are the probabilities our model calculates for the baserunner and hitter to reach their targeted base(s). The blue dots represent the fielders, and the red dots represent the hitter and the baserunner. The red and blue dashed lines indicate the approximate basepath route of the baserunner and hitter, respectively, and the purple dotted line represents the approximate ball trajectory. [Link](#) to play.

3.2 Determining Optimal Cutoff Actions

Now that we had the safe probabilities for every baserunner in our dataset, we could determine the cutoff man's optimal action. We considered three options:

1. "Continue the throw": The cutoff man throws to the base the outfielder is targeting. In the UCSD play, this would be the case that the pitcher throws home.
2. "Cut and relay": The cutoff man throws to a different base. In the UCSD play, this would be the case that the pitcher throws it to 3rd base.
3. "Cut and hold": The cutoff man holds on to the ball. In the UCSD play, this is what the pitcher ended up doing.

Note that we assume all throws from the outfield can be cut off by an infielder. In reality, this may not be the case, as outfielders trying to throw directly to home might throw the ball too high for someone to cut it off.

Using a combination of the RE24 run expectancy data and our safe probabilities, for each play, we calculated the expected run values (xRVs) for the three potential cutoff actions (*Appendix 4*). *Figure 5* shows an example of our process. After comparing each of the expected run values, the cutoff action with the lowest xRV is deemed *optimal*.

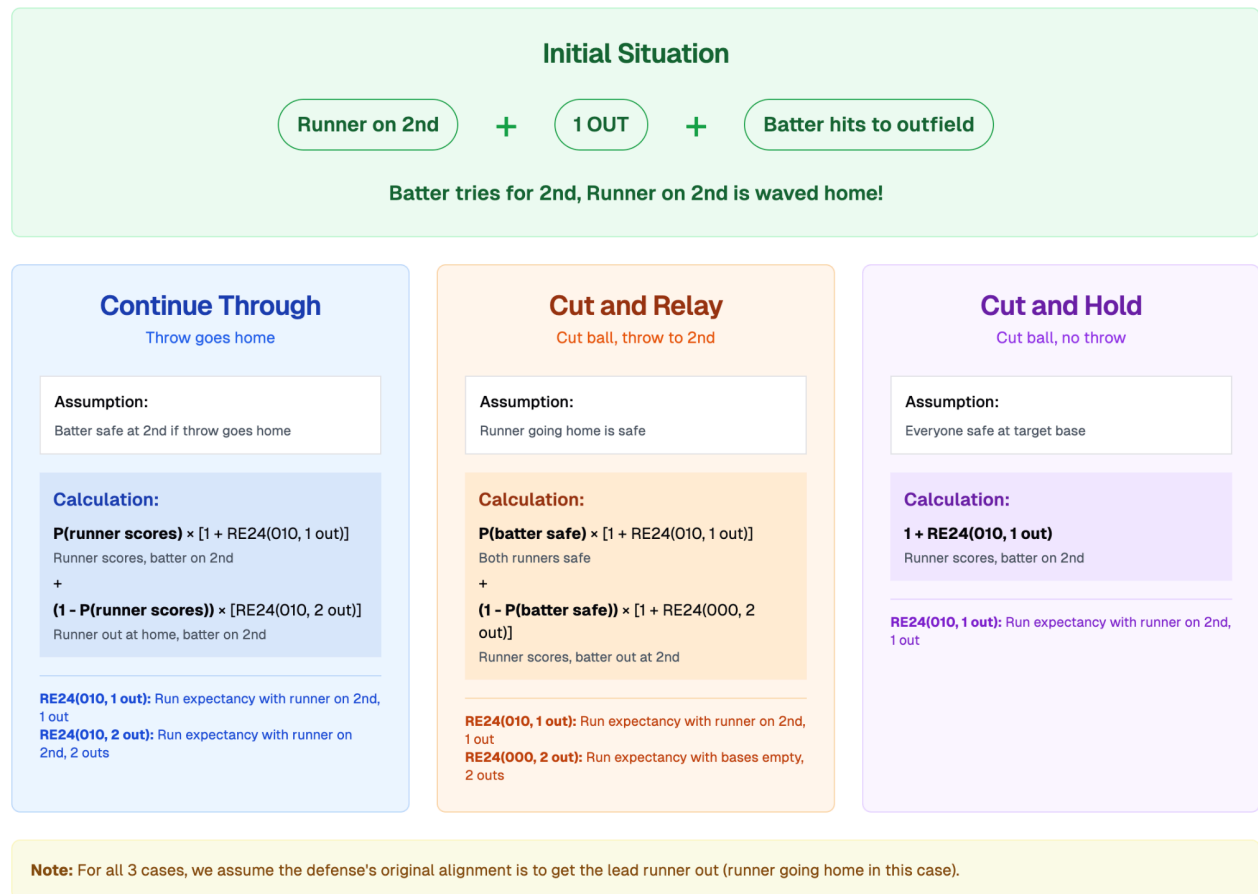


Figure 5: Example situation is runner on 2nd & 1 out. The graphic outlines expected run value calculations, using our derived safe probabilities from the initial model, for each possible cutoff action.

A high percentage of plays showed close differences in each cutoff action's EVs, so to account for "cut and hold" never having the lowest EV, we set a run value threshold of 0.08 that would default to "cut and hold". If continuing the throw or cutting-and-relaying were within 0.08 runs holding the ball, we assumed the fielder should cut and hold to minimize a possible error (*Appendix 4*). See *Figure 6* for how the optimal cutoff actions are distributed through cutoff decision zones and *Figure 7* for a play where "cut and hold" is chosen as the optimal action.

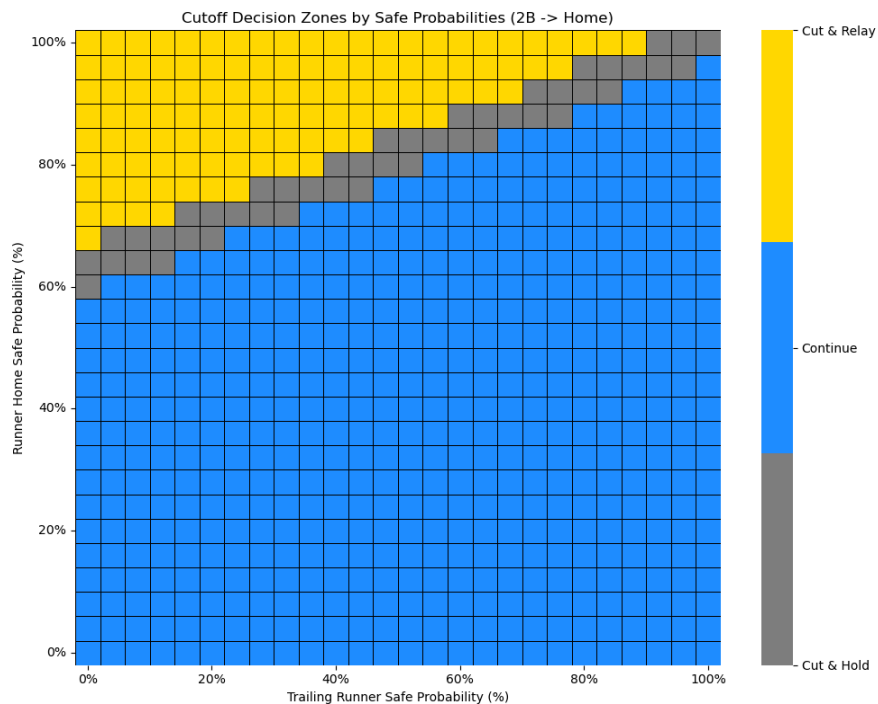


Figure 6: For a base path such as 2B $\rightarrow$ home, the probabilities of the runner being safe at home, and the trailing runner (batter, in this case) being safe at second, result in the optimal cutoff action.

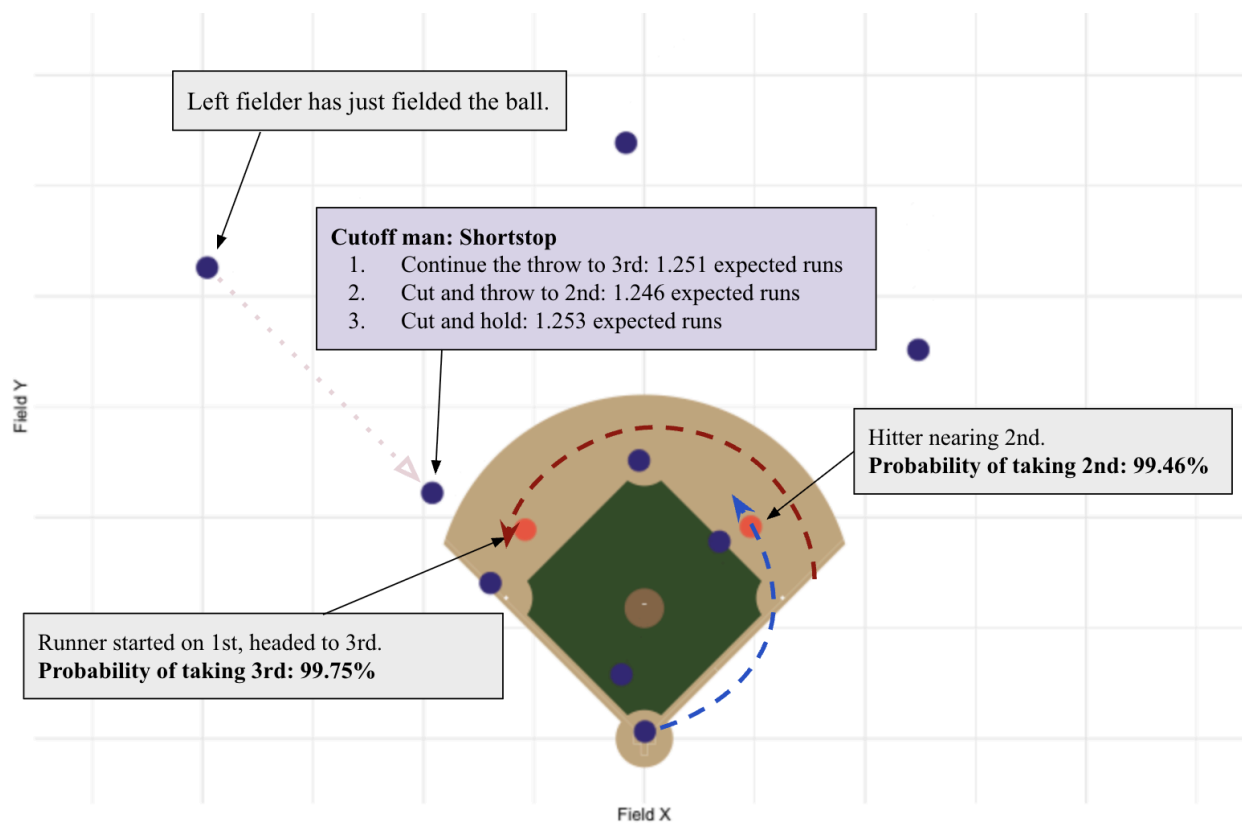


Figure 7: **Click** on the image to see an animation of the play. This play's optimal cutoff action is to "cut and hold", as the expected run values are all within 0.08 of each other. Both runners are very close to reaching their target base, so the cutoff man holding the ball would prevent any errors and inadvertent extra bases. [Link](#) to play.

Additionally, deriving the number of outs on any given play was a key aspect to improving this logic (*Appendix 1*). You can see the optimal cutoff action distribution by out in *Figure 8*.

Figure 8: Optimal cutoff action distribution per outs on the current play. Understandably, there are fewer plays with 0 outs because fewer runners are on base with 0 outs, leading to fewer possible cutoff plays as a whole. We can see that a higher percentage of plays with 2 outs are recommended to “continue through”, because getting an out is more valuable in those scenarios.

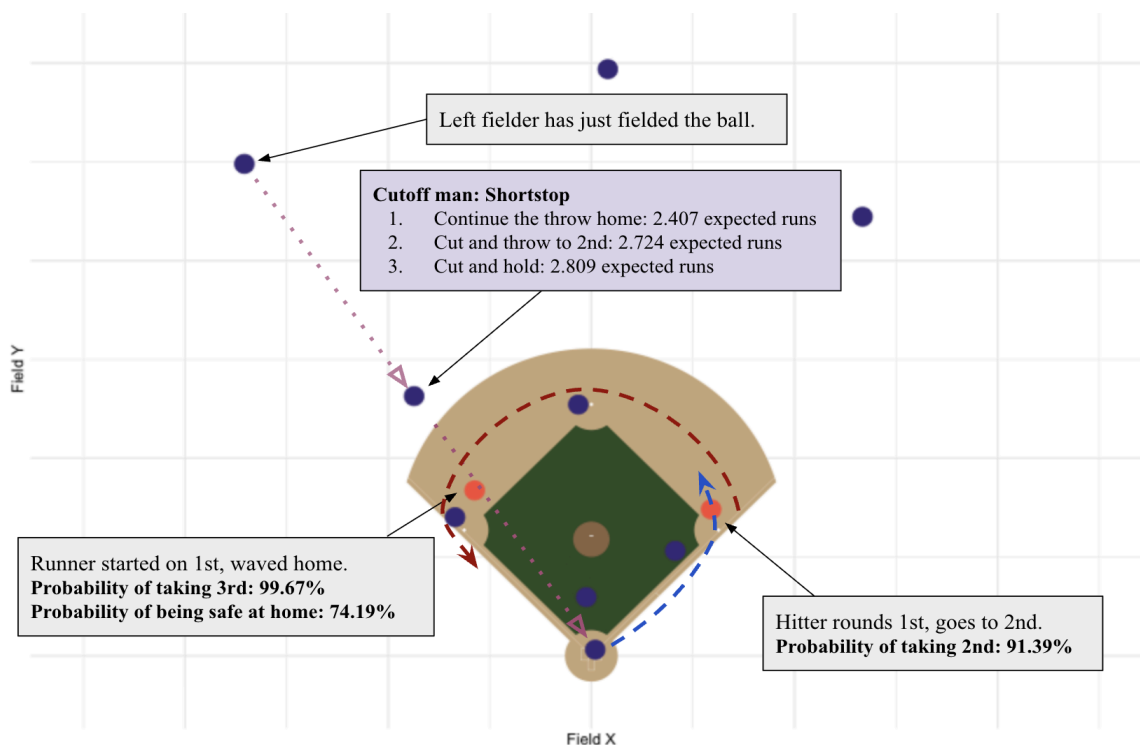
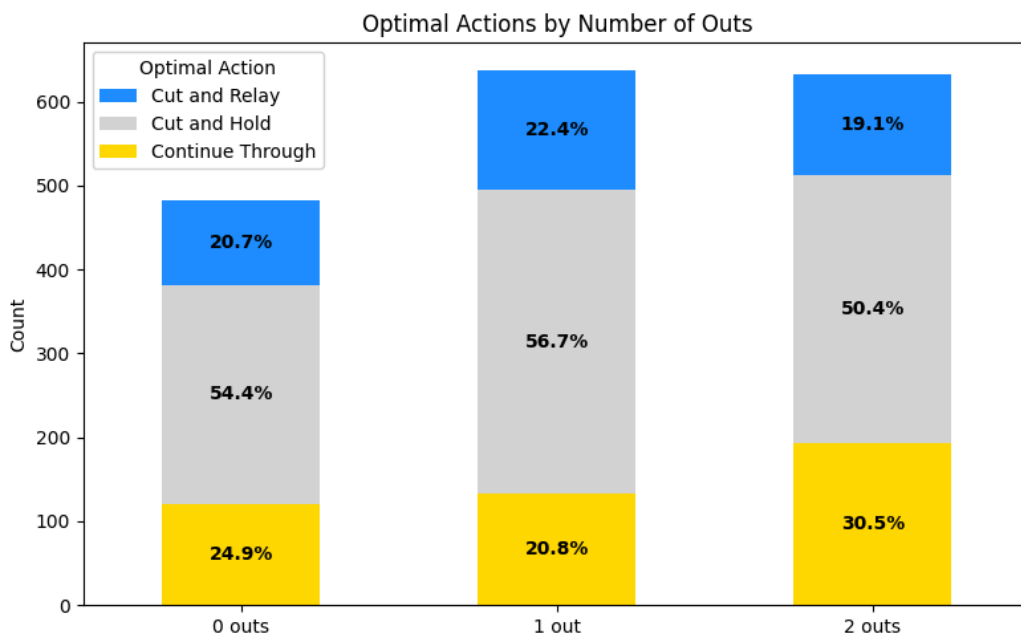


Figure 9: Same play as *Figure 4*. The optimal action for the cutoff man is to continue the throw home, as that yields an expected 0.317 runs less than relaying it to 2nd. In reality, the cutoff man ended up doing that, and got the runner out at the plate!

3.3 Optimal Action Model

The optimal actions we determined in section 3.2 effectively communicate the correct decision making to the infielders in *hindsight*. But, we were missing the “why”. We wanted to build a model that could tell a cutoff man, “The outfielder has the ball and is throwing it my way. What should I do in real time?”

We made sure only to include information that the infielder would be able to pick up in real time (like approximate distances), or metrics that could be listed in a pre-game scouting report (like fielder arm strengths and runner speeds). Using the optimal cutoff actions as a target variable, we trained a Random Forest Classifier model with the following features:

- fielder distance to the targeted base;
- baserunner distance to the targeted base;
- cutoff man distance to targeted base;
- outfielder arm strength;
- baserunner sprint speed;
- number of outs;
- baserunning path (1st to 3rd, 2nd to home, etc.)

Figure 10 showcases the importance of the features in the model.

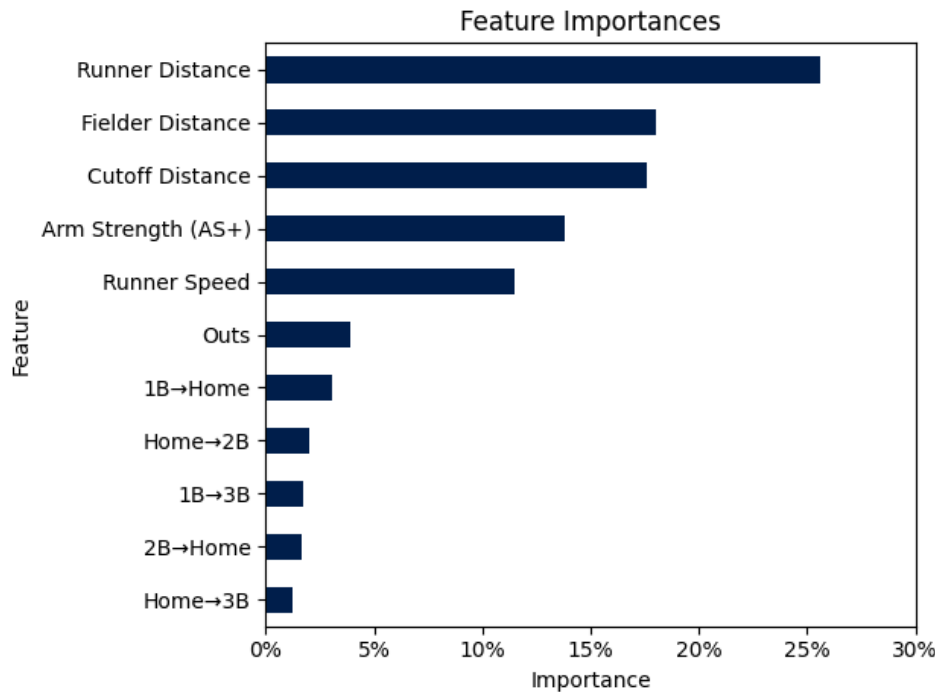


Figure 10: All features used in the larger cutoff model and their relative weight in the random forest classifier. As you can see, the runner’s distance to the base, followed by the fielder and cutoff man’s distance to the base the most critical to the model’s decision, followed by arm strength and runner speed. We intuitively agree!

Overall, this model was fairly successful at predicting the optimal cutoff action based on the features, with about a 77% accuracy rate across all three cases. The full accuracy report is shown in *figure 11*, and the decision matrix in *figure 12*. Considering, without a model, choosing the optimal cutoff action at random would yield 33% accuracy and choosing only “cut and hold” would yield ~50% accuracy, we feel confident in our model’s results.

Model Performance by Action

	PRECISION	RECALL	F1 SCORE
Cut and Hold	0.80	0.86	0.83
Cut and Relay	0.69	0.75	0.72
Continue	0.77	0.61	0.68
Overall Accuracy			0.77

Figure 11: Accuracy report for random forest classifier model. The model can predict the optimal cutoff action 77% of the time using the distance, arm strength, and runner speed features.

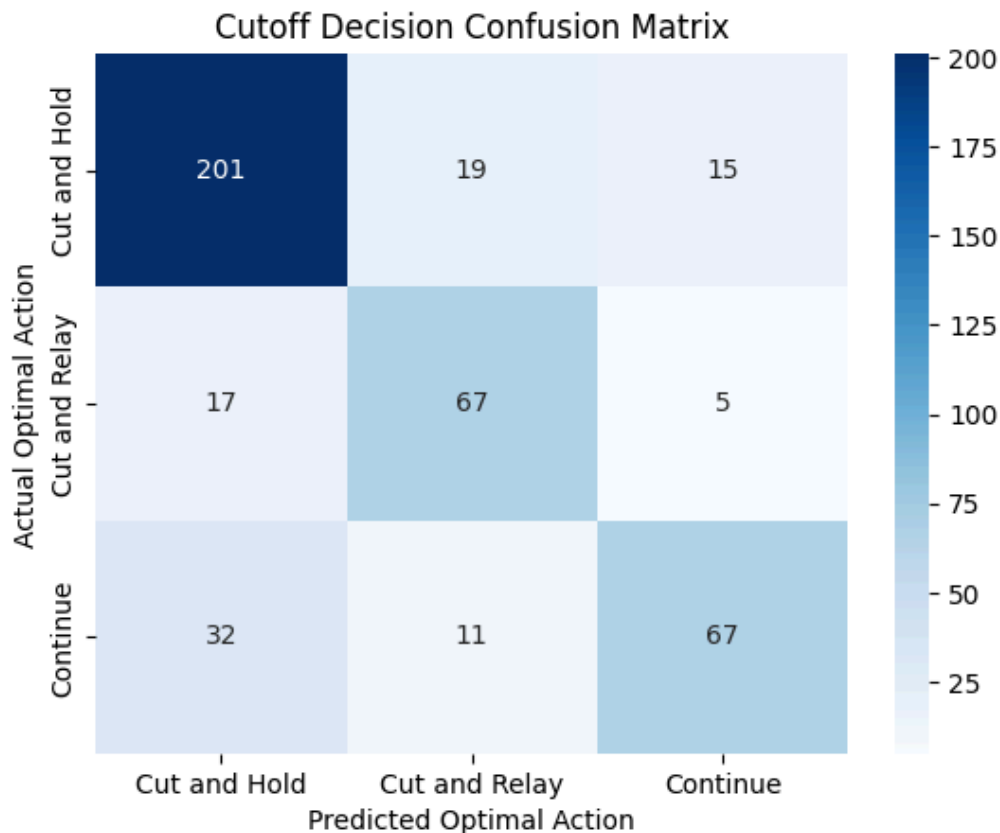


Figure 12: Confusion matrix for model predicted optimal cutoff actions vs. actual optimal actions. This serves as a representation of the model’s accuracy.

4. Takeaways & Analysis

Our analysis compares our model-predicted optimal cutoff actions with fielder observed cutoff actions in order to provide both big picture strategy insights as well as team and fielder-specific scouting takeaways. The team and player-level takeaways are supplemented by an interactive dashboard we developed to communicate our cutoff decision-making analysis.

4.1 Model vs. Observed: Are Cutoff Men Making the Right Decisions?

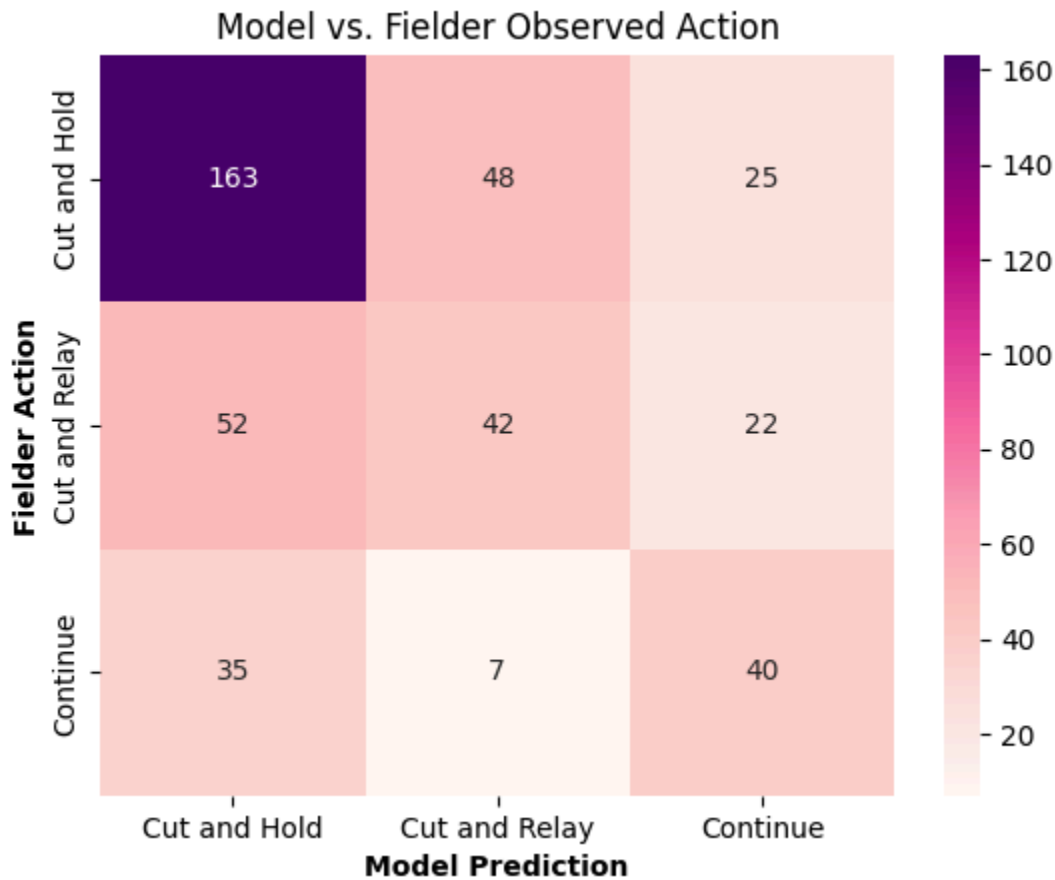


Figure 14: Confusion matrix comparing random forest model's predicted optimal cutoff action vs. observed fielder action. The smaller sample is due to a 75/25 training/testing split².

Figure 14 represents a confusion matrix between the *model-predicted* and *observed* cutoff actions. It is important to note that this is different from the confusion matrix in section 3.3 (figure 12), which compared *model-predicted* cutoff actions with *optimal* cutoff actions. In short, the matrix in this section lets us analyze fielder decision-making, while the previous one helped

² A modeling technique where the model is trained on 75% of the data, randomized, and then tested on the remaining 25%. This prevents overfitting.

us analyze the model's accuracy. Comparing model-predicted actions with observed actions allows us to fairly and realistically evaluate whether fielders made the correct decision based on the information available to them during the play.

One clear trend emerges from the matrix: fielders are not cutting and holding the ball nearly as often as they should. Of the 250 plays where the model predicted that the fielder should cut and hold the ball, the fielder actually cut and held it in only 163 of them. This indicates a widespread room for improvement for cutoff men, specifically related to over-aggressiveness. Rather than optimizing for safety and holding the ball, fielders appear to be opting for riskier decisions to try and get runners out, costing their team significant run value in the long term.

4.2 Dashboard

Explore the dashboard here: <https://maninthemiddle.shinyapps.io/smtcutoffdashboard/>

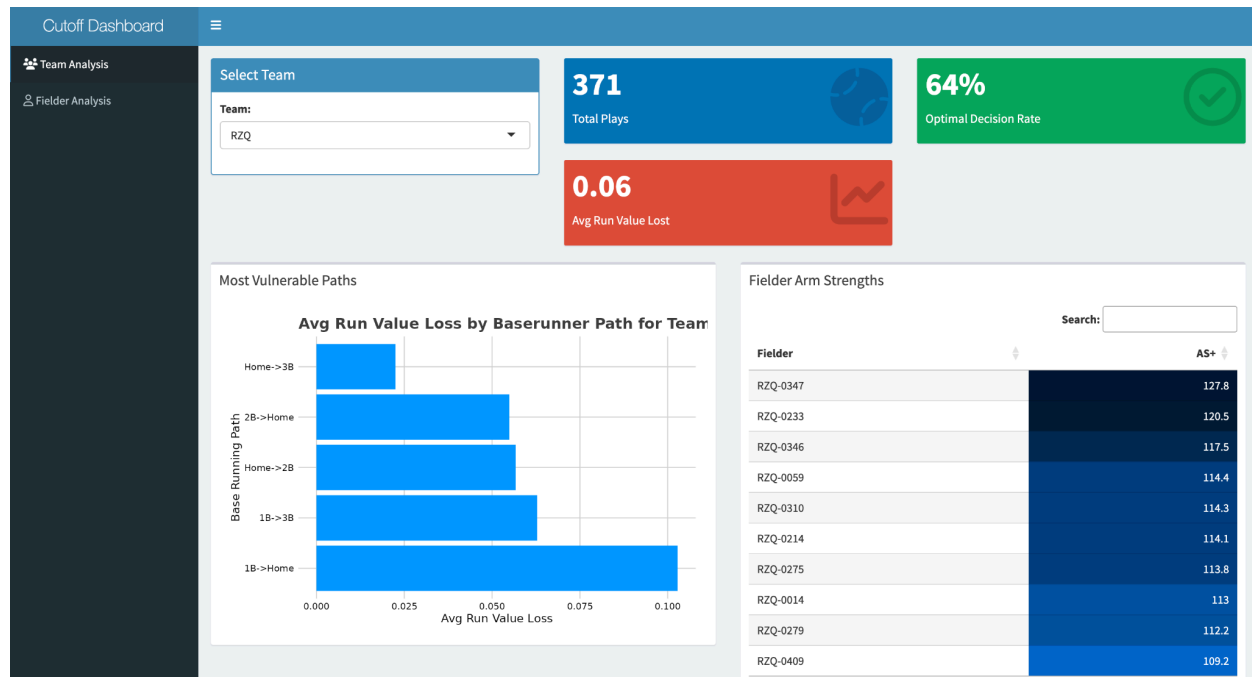


Figure 16a: An example of what the team tab of the dashboard looks like. You can select a team, and get a breakdown of their performance on cutoff plays, as well as their most vulnerable baserunning paths and their outfielders' arm strengths.

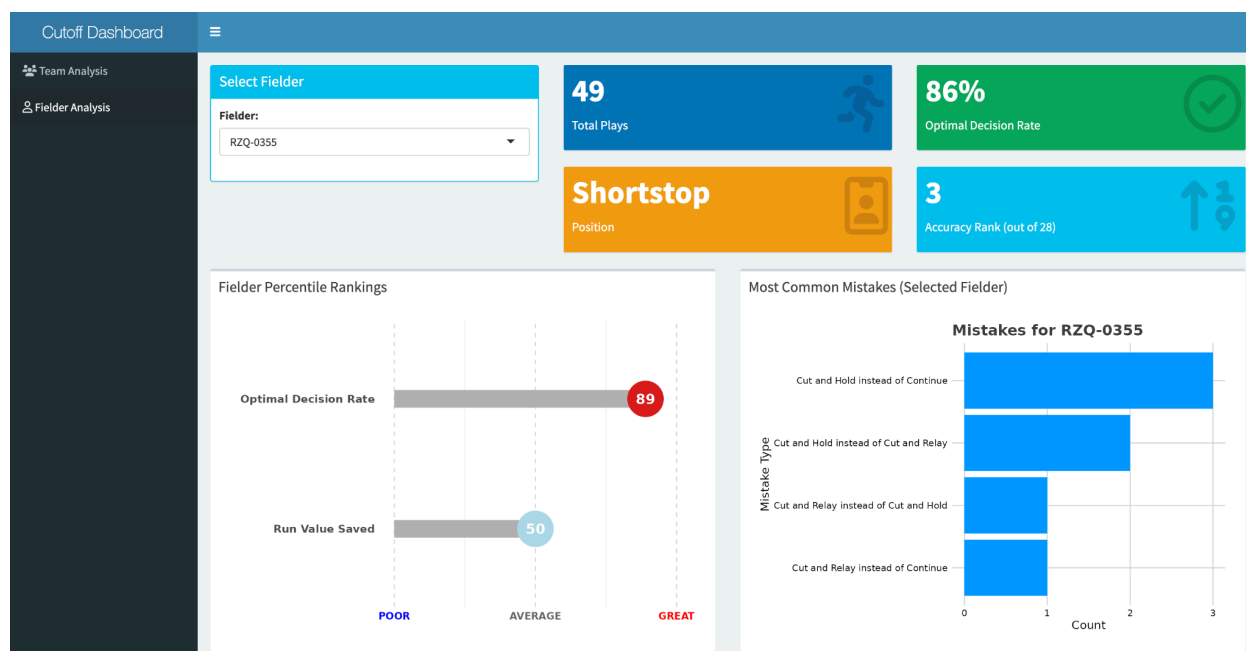


Figure 16b: Dashboard fielder tab view. You can select a player, and get a breakdown of their performance on cutoff plays, including their accuracy and most common mistakes.

In order to make these insights more digestible, we created an interactive dashboard using R Shiny³ (figure 16a and 16b). With this dashboard, you can toggle between specific teams and fielders to understand their overall decision-making skills, as well as identify particular areas of weakness. With several season's worth of data, we foresee a tool like this being incorporated into a player or team's Baseball Savant or Fangraphs page to round out their fielding evaluations.

The dashboard communicates several insights that reinforce our belief that teams should be cutting and holding the ball. For instance, across the board, teams were most vulnerable on plays with a runner going from 1st base to home, losing an average of 0.1 expected runs on those plays. This is likely explained by over-aggressive cutoff men trying to get a runner out at the plate when they are almost certainly safe.

The dashboard also serves as a valuable scouting tool. Teams can coach their baserunners to be more or less aggressive on the basepaths, based on the decision-making habits of opposing cutoff men. For example, if the shortstop throws the ball home a lot more than he should, then teams should encourage their trailing baserunners to try and advance more (i.e. from 2nd to 3rd).

The main limitation of the dashboard right now is the lack of a strong sample size. With no individual player logging more than 50 plays, it is difficult to generalize these results to a higher level takeaway about that player's actions. However, if built with the high-quality data that professional teams have access to, we believe this tool has the potential to reshape how

³ R Shiny is a readily available package in R that facilitates building interactive applications.

teams evaluate and approach cutoff play decision-making, and allow them to exploit tendencies in high-pressure moments.

5. Conclusion

This project provides a novel approach to evaluate one of the most under-analyzed components of baseball: cutoff play decision-making. We were able to simulate, evaluate, and model optimal cutoff decisions across hundreds of plays, allowing us to assess not only whether a fielder made a correct decision in hindsight, but also what they should do in real time during a play.

The big picture takeaway is that cutoff men are too eager to target the leading base runner, and should be holding onto the ball more than they are. But, that's just the bird's eye view. Our model and analysis is most valuable to coaches and players on a team and fielder level. It provides them with accurate insights into an opponent's cutoff decision making habits, coupled with the opportunity to smoothly test different scenarios and draw conclusions with our interactive dashboard.

However, several limitations certainly do still remain. The most pressing challenge was data quality. The lack of play-by-play and game state information in the data forced us to define our own logic to determine if players were safe or out, which could cause inaccuracy in the initial safe probability model. The limited number of recorded cutoff plays also restricted the sample size for player-level conclusions. In addition, we assumed that all throws could be cut off and that players had complete awareness of the play, which may not always reflect the reality in baseball. Finally, while our method of aligning the optimal and observed distributions may work in the short term, future work could incorporate historical error likelihoods or game-theory based models to replace this methodology with a more dynamic and context-aware adjustment.

Looking ahead, there are several ways to expand our project to make it more impactful. Scaling up our methodology with MLB or MiLB quality data would improve the accuracy and scalability of our results from end to end. Future versions of our dashboard could also integrate qualitative scouting, such as player tendencies or coaching tips, and compare specific runner-fielder situations over a large sample size. Ultimately, our hope is that this work leads to further exploration into cutoff play decision-making. Maybe the smartest throw in baseball is the one that the man in the middle never makes.

6. References

We drew inspiration from Billy Fryer's [2022 SMT Data Challenge Project](#) to come up with our expected run value formula based on safe probabilities and the RE24 matrix.

7. Acknowledgements

A huge thank you to SMT's Dr. Meredith Wills and Billy Fryer for organizing this data challenge and supporting us throughout the whole process. Also, thank you to Adam Esquer, VP of Research and Development for the San Diego Padres, and Ajay Patel, Junior Data Scientist for the Pittsburgh Pirates, for offering us advice on our optimal action determination logic and giving us feedback on the paper. We couldn't have completed this project without their support!

8. Appendix

Appendix 1 - Baserunner Safe/Out Criteria

As previously mentioned in sections 2.1 and 5, safe or out results of a play were not given. This was a major problem, as our safe probability model needed to be built on the binary decision of whether the runner was safe or out.

Before we began the process of labeling runners safe or out, we needed to filter for instances where the runner attempted to take an extra base. We established a threshold about halfway down each baseline where if the runner was to cross this threshold, we deemed that they were "attempting" to reach the next base. This would exclude instances where the batter takes a big turn around first on a base hit and scurries back to the bag, or where an overly aggressive runner on 2nd base rounds 3rd on a base hit, but is held up by the third base coach.

We began the process of finding safe/out instances by taking a deeper look into who was on base the following play. This was an easy way to tell if the runner was safe at their targeted base, barring extreme or rare scenarios. In our data, we created new binary columns to indicate whether or not the baserunner appeared at their targeted base during the next play (subsequent `play_id`). This was the most concrete data that we had for labeling a runner as safe. We couldn't rely on this "play after" philosophy 100%, though. The dataset was messy with various null instances through the dataset. Additionally, this was only applicable to plays at second or third, failing to account for runners at home plate.

To assist our "play after" philosophy, we created several boundaries and rules using baserunner and fielder tracking data to confidently label a runner safe or out. This process was backed by manually checking dozens of animated plays. At the moment the fielder at the targeted base (e.g. third baseman if third base was targeted) acquired the ball, we calculated the distance of the fielder and the baserunner to the base. The following requirements were to be met to deem a runner safe at the moment the fielder acquired the ball:

- a. the runner was less than five feet away from the base and;

- b. the runner was at least one foot closer to the base than the fielder

or...

- c. the fielder was more than nine feet away from the base.

In combination with this methodology, for plays at second or third, if the runner appeared at the targeted base in the following play's data, we would force the runner safe despite any contradicting estimate using our set of arbitrary rules. *Figure 17* illustrates a promising pattern in our safe/out decision making process.

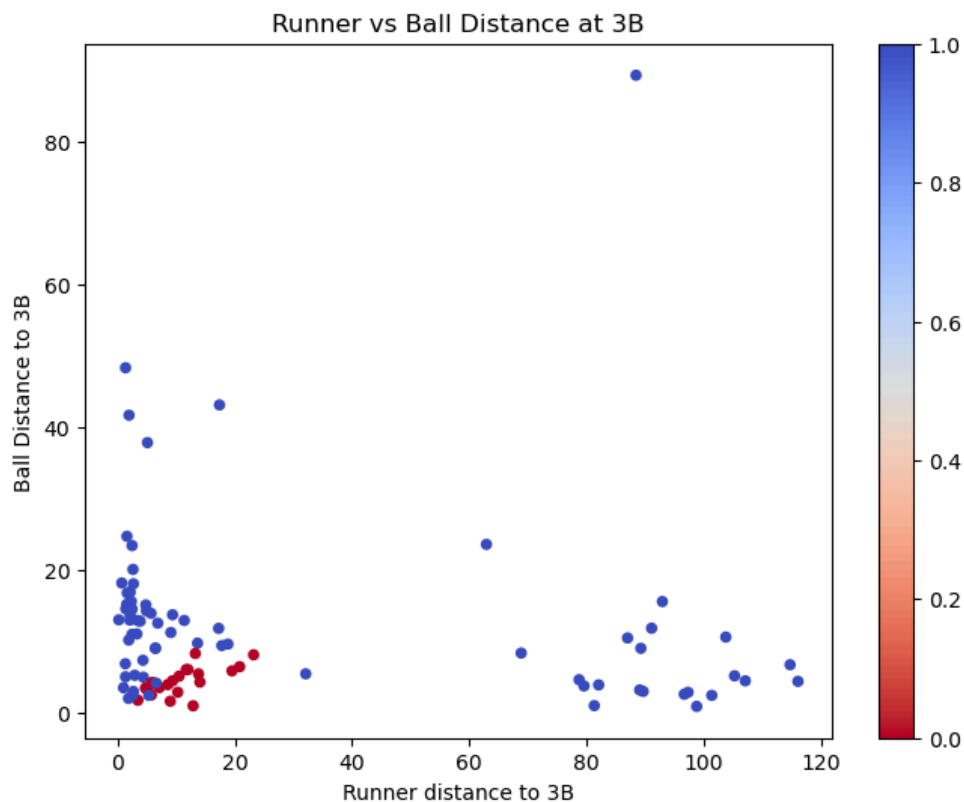


Figure 17: Scatter plot of runner and ball distance to third base for runners going from 1st to 3rd. Blue represents a runner is safe, and red represents a runner out. The graph backs our safe/out methodology, showing a clear pattern in the data.

Using our safe/out methodology, the brier scores are generally better in the models where runners are targeting second or third (showcased in *figure 2*). This is because in the data, we can't see if a runner reached home by checking the base state of the following play, so basepaths $2B \rightarrow \text{Home}$ and $1B \rightarrow \text{Home}$ relied on our arbitrary set of rules.

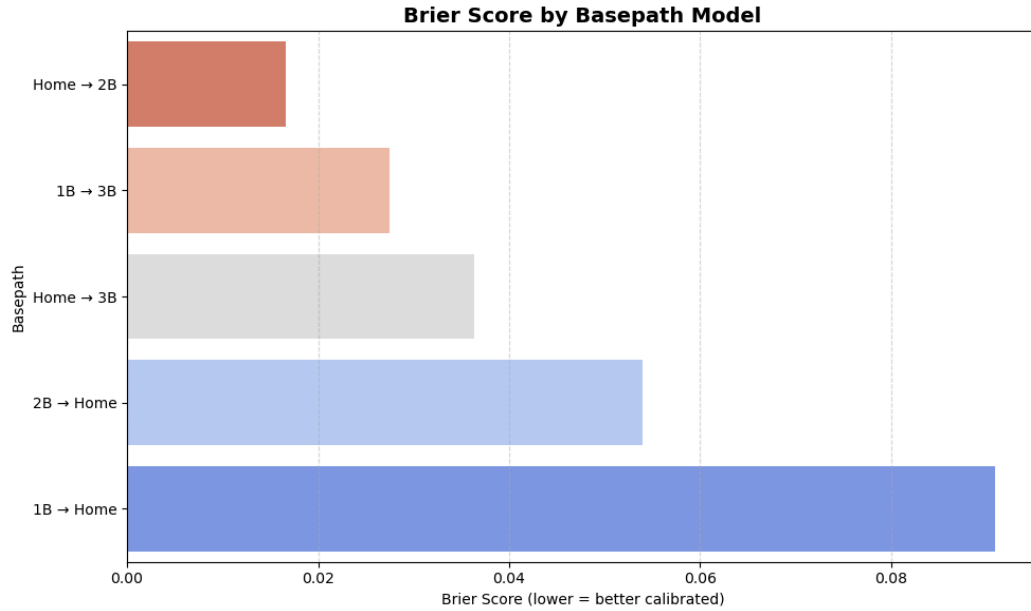


Figure 2: Seen first in section 3.1

It was important to devise a strong procedure to identify whether a runner was safe or out at their targeted base. Through the two steps outlined in this appendix, we feel confident in the accuracy of our labeling. Of course, if the dataset included those labels to begin with, we would be more confident with the data's accuracy, and therefore our initial safe probability model's accuracy, and would've been able to focus our attention elsewhere. Yet, we are happy and satisfied with our methodology given the brier scores for each of our five safe probability models.

Appendix 2 - Determining the Number of Outs

As previously mentioned, the dataset did not include any information about the number of outs at any given time. However, we were given data that, when combined with knowledge about different ways that outs can occur, allowed us to identify the vast majority of outs as they occurred.

Percentage of outs at the time of the pitch	
0	35.03%
1	33.15%
2	31.81%
3	0.02%

There were two extremely helpful data points given that helped derive outs. First, we could see whether each base was occupied or not at every frame. Additionally, this data included codes that uniquely identified each player, allowing us to track individuals around the bases. Second, the type of play occurring at each time stamp allowed for knowledge of what was going on when runners advanced bases or disappeared. Combining these two data points with general baseball knowledge of how outs occur, we created rules to label outs and track the number of outs over time in each half inning.

Included in the type of play tab was an indicator if there was a home run, so for the purpose of our outs calculations, we skipped home run plays entirely. Next, we created the following four rules to identify outs.

1. Batter hitting the ball into play and not reaching base
 - a. I'll explain each more specifically
2. Batter seeing at least three pitches and not reaching base
3. Pickoff throw to base and runner is no longer on base
4. Batter reached first base and runner on first was no longer on base

Given the number of plays and innings in the entire dataset, there should have been __ total outs. This methodology found ___ of them. The discrepancy is partially explained by “missing outs,” those we can not identify by regular logic. Also, there could have been games that contained fewer outs than expected due to a walkoff or weather ending it early.

This methodology is not perfect, as there is a near-infinite number of ways runners can be out in non-force plays, but we were able to identify (x%) of all outs, and this distribution (attach) indicates that our logic mostly worked.

Appendix 3 - Optimal Cutoff Action Determination

The exact mathematical formula we used to determine the optimal cutoff actions is shown below:

Let $A = \{a_1, a_2, \dots, a_n\}$ be the set of attempted base advances on a play. Each $a_i \in A$ has a probability $p_i = \Pr(\text{Success of } a_i)$. We compute the **expected run value** (xRV) by summing over all 2^n possible combinations of outcomes. Each runner attempt can independently result in either a success (safe) or a failure (out), which is what leads to 2^n total binary outcome vectors.

$$\text{xRV} = \sum_{\mathbf{o} \in \{0,1\}^n} \left[\left(\prod_{i=1}^n p_i^{o_i} (1 - p_i)^{1-o_i} \right) \cdot (\text{RE}_{\mathbf{o}} + \text{RS}_{\mathbf{o}}) \right]$$

where:

- $\mathbf{o} = (o_1, o_2, \dots, o_n)$ is a binary outcome vector (1 = safe, 0 = out)
- $\text{RS}_{\mathbf{o}}$ = number of runners that successfully scored (i.e., advanced to Home)
- $\text{RE}_{\mathbf{o}}$ = run expectancy value from the RE24 table for the resulting base state

After coming up with these values, we set the action with the smallest xRV to be the optimal action. A key part of this process was determining a threshold value to default the optimal cutoff action to “cut and hold”. This was required because based on our mathematical process using probabilities, the expected run value of continuing or relaying would always be greater than cutting and holding.

In order to arrive at this threshold number, we first generated the *observed* distribution of cutoff actions. In other words, what cutoff men actually did in our data. We used the following logic to assign each play with an observed label:

- “Continue”: The ball was thrown from the outfield to a cutoff man, and the cutoff man subsequently threw the ball to the player assigned to the target base of the leading runner (Catcher for a runner going Home, 3rd base for a runner going to 3rd).
- “Cut and relay”: The ball was thrown from the outfield to a cutoff man, and the cutoff man subsequently threw the ball to the player assigned to the target base of a trailing runner.
- “Cut and hold”: The ball was thrown from the outfield to a cutoff man, who did not make any subsequent throws. Note that if the outfielder threw it directly to the 3rd baseman with a runner approaching 3rd, we considered this a “continue” and not a “cut and hold” since the intent was to get the runner out.

The observed distribution is shown below:

Action	Count
Cut and Hold	530
Continue	177
Cut and Relay	92

After determining the observed distribution, we grid searched through various threshold values, measuring the distance between the resulting optimal cutoff distribution and the observed one. We then picked the threshold value that made the optimal distribution look closest to the observed one. With more time and higher quality data, we would definitely try to improve this process to more accurately assess the point at which cutting and holding is optimal, but we thought that this grid searching process was suitable for our purposes, and generally aligned with our intuitive thoughts on the correct decision through manual validation of the plays.

Appendix 4 - Feature Engineering

We outlined in section 3.1 the features that we used for our initial safe probability model, and in section 3.3, the features that we used for our larger, optimal cutoff action model.

We want to emphasize that distance-related features were calculated at the time the outfielder acquired the ball. We felt the moment this occurred was a generally good indicator of when the cutoff man has to start taking in all the developing aspects of the play to make his cutoff decision.

The distance-related features are all calculated very similarly – the distances from the fielder, baserunner, and cutoff man distance to the baserunner’s targeted base. These were calculated by taking the euclidean distance⁴ from the player to the base, on the xy-plane.

The outfielder arm strength and sprint speed features were calculated a little bit differently. These were individual metrics at their core. Each player and runner ID represented a unique MiLB player with a unique arm strength or speed on the basepaths.

To calculate arm strength, we used a similar process to calculating velocity out of a pitcher’s hand. We identified the two closest timestamps after the outfielder released the ball, calculated the euclidean distance between the ball at those two timestamps, and divided by the difference in the timestamps. Using this method was superior to calculating velocity by taking (total distance / total time), because the latter left room for air resistance, wind, and the possibility of the ball bouncing and slowing down. We felt our calculation was a more accurate

⁴ distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

representation of true individual arm strength. After that, we converted the metric to miles per hour for ease of comprehension.

We saw many instances where there was no competitive play at a base, meaning the throw from the outfielder into the infield was a nonurgent lob. For instance, if the outfielder saw that a runner rounding third and heading for home was already $\frac{3}{4}$ down the basepath, they might gently toss the ball to the cutoff man, aware of the insanely small chance their throw would beat the runner to home plate. To avoid taking this into consideration, we used the max arm strength for each fielder ID as their true arm strength. But, this didn't solve our problem just yet. There were several fielders who we only had one or two play's worth of data for, so if their throws were uncompetitive, their max arm strength would be noticeably low. To work around this, if their max arm strength in the dataset was less than 50 miles per hour, we replaced their arm strength metric with the mean arm strength for their position. This raised the average arm strength in the dataset from 57.6 to 68.1 mph, a more accurate representation holistically.

Finally, we standardized the arm strength metric to 100, similar to how OPS+ and ERA+ are calculated. We aim for our analysis to be replicated and used as an advanced scouting tool for teams, so using a standardized metric can help with coach or player understanding. The metric is called AS+. If a fielder's arm strength is 20% weaker than the average, their AS+ would be ~80, and vice versa, if their arm strength is 20% stronger than the average, their AS+ would be ~120.

Lastly, we needed to calculate sprint speed for unique baserunner ID's. Our philosophy for finding an accurate sprint speed was slightly different from arm strength. Instead of taking the max sprint speed, we felt that taking each baserunner's 90th percentile sprint speed would more accurately represent their speed in most plays. To do so, we looped through each baserunner's running paths to find their fastest 1-second interval. Keeping the data clean, we discarded any instances where the runner ID was null. Next, we calculated for the 90th percentile sprint speed, only keeping runner IDs that appeared in five or more plays to ensure a decent sample size.

In *Appendix 1*, we outline our process for identifying the number of outs and how it shaped the model's performance, our findings, and allows for deeper analysis. We used outs as part of our optimal cutoff action logic because we feel this is a notable variable in any high-pressure cutoff play.

If we look deeper into the expected run value formula, we can see the true effects that the number of outs has on a cutoff decision. For this example, let's assume the following situation and relevant safe probabilities:

- runner on 2nd base, base hit to centerfield,

- runner on 2nd base has a safe probability at home of 90%,
- trailing runner has a safe probability at 2nd of 80%,
- 1 out,
- calculating for expected runs where the cutoff man is “continuing” the throw,

$$xRV_{\text{continue}} = P_{\text{lead_runner_out}} \cdot (RE24_{2\text{outs, runner on 2nd}}) + P_{\text{lead_runner_safe}} \cdot (RE24_{1\text{ out, runner on 2nd}}) + 1$$

This equation represents the expected run value (EV) if the cutoff man decides to continue the throw home. With one out, the equation considers both scenarios—where the runner at home is safe and out.

Let’s consider the same situation with two outs.

$$xRV_{\text{continue}} = P_{\text{lead_runner_out}} \cdot (0) + P_{\text{lead_runner_safe}} \cdot (RE24_{2\text{ out, runner on 2nd}}) + 1$$

The equation reflects that if the cutoff man throws the lead runner attempting home out, there’s no future base-out state to consider because the inning is now over. So, the probability that the cutoff man throws the lead runner out at home has a much bigger impact on the expected runs. With two outs, the expected run value of continuing the throw is more likely to be less than the expected run value of cutting and relaying to another base. For this reason, it was crucial we considered outs as a feature in our final optimal cutoff action model.

Appendix 5 - Modeling Details

We developed two different models as part of our pipeline: A safe probability model and an optimal action model.

Our safe probability model was a logistic regression model modeled from the point the outfielder received the ball. We used two features, runner distance to target base and outfielder distance to target. We chose to use a logistic regression model because of its simplicity and interpretability for binary classification. The logistic model We used an 80/20 train/test split on the data, and received strong results without adding unnecessary complexity.

Our optimal action model was a random forest classification model trained on 11 input features. We chose a random forest model because it is adept at handling nonlinear feature interactions, and is robust to overfitting. We used a stratified 75/25 train/test split followed by 5-fold cross-validation on the training set. This allowed us to evaluate the model’s performance across multiple different partitions within the training data, helping it reduce variance and avoid overfitting to a small subset of plays. We one-hot encoded basepaths to make them numerical features, telling the model which specific baserunning path the runner was on during a play.

Since our target variable (the optimal cutoff action), was imbalanced, with a higher proportion of “cut and hold” decisions, we used SMOTE (Synthetic Minority Over-sampling Technique) to address this class imbalance. SMOTE generates synthetic samples of the minority classes (“cut and relay” and “continue”) until they are balanced, allowing the model to more clearly distinguish between the different actions.

Finally, we performed a grid search over various different hyperparameters to find the ideal set. The metric for evaluating for this hyperparameter tuning was cross-validation accuracy, and resulted in the following configuration:

Hyperparameter	Value	Explanation
n_estimators	100	Number of trees.
max_depth	20	Maximum depth of the trees.
min_samples_split	2	Minimum samples required to split a node.
min_samples_leaf	1	Minimum samples required at a leaf node.

For this final model, we also tried using an XGBoost modeling approach. However, the results for the XGBoost model were fairly similar to the Random Forest model, so we opted to use the Random Forest for the sake of clarity and ease of interpretation. The results of the XGBoost model are shown below.

In the future, we could improve this model by considering features like ball trajectory and additional game context (inning, score, etc.). We could also explore strategies such as reinforcement learning to better model the sequential decision-making of cutoff plays.